



Ayush Goel

Roll No: 23CH10092

B.Tech (Hons.) in Chemical Engineering

Micro Spl. in Artificial Intelligence and Applications

Indian Institute of Technology, Kharagpur

+91-8284971401

ayushgoel2005@kgpian.iitkgp.ac.in

goel85360@gmail.com

GitHub Profile

LinkedIn Profile

EDUCATION

Indian Institute of Technology, Kharagpur

Bachelor of Technology (Honours) in Chemical Engineering

2027

CGPA: 9.04

Delhi Public School, Patiala

2023

All India Senior School Certificate Examination (AISSCE), CBSE

Percentage: 93.4

Ryan International School, Patiala

2021

All India Secondary School Examination (AISSE), CBSE

Percentage: 95.4

RESEARCH AND INDUSTRY EXPERIENCE

Machine Learning Intern @ Chi SquareX | Remote

Dec 2024 – Jan 2025

- Built an RL-compatible trading environment using **OpenAI Gymnasium**, with a well-defined state and action space for the RL agent
- Trained and evaluated a reinforcement learning model using a **reward structure** based on profit/loss, applying **PPO** and **A2C** algorithms
- Developed the **Call Spread Butterfly** strategy with an **object-oriented** design to simulate option contracts based on the agent's policy
- Designed an interactive and robust GUI in **Streamlit** to monitor, train, and validate the agent's performance in diverse market scenarios

Undergraduate Researcher @ Aerial Robotics Research Group, Prof. Somesh Kumar | IIT Kharagpur

May 2024 – Sep 2025

- Engineered real-time human detection using **MediaPipe FaceMesh**, enabling fast, accurate facial landmark tracking on aerial footage
- Integrated a pre-trained **YOLOv8** model to effectively detect targets and obstacles in real-time for autonomous drone perception tasks
- Built an **Iris drone simulation** in Gazebo using MAVROS velocity topics for smooth and precise manual keyboard-controlled navigation
- Designed customized gradient-based edge detection and **Hough Line Transform** algorithms for perception in challenging environments

PROJECTS

Intelligent Path Planning in 2D Robotic Environments | Self Project

May 2025 – Jun 2025

- Implemented the **Probabilistic Roadmap** Algorithm manually for collision-free motion planning using a **2D** binary image maze system
- Extracted obstacle boundaries and free-space regions to construct accurate configuration nodes with user-defined, flexible sample sizes
- Built node connections using the **KD-Tree** neighbor querying with up to **200** neighbors per node, for denser and reliable roadmap links
- Applied the **A* Algorithm** with a **heuristic** and **priority queue** to locate one of the shortest trajectories in the PRM-generated roadmap

Rubik's Cube Solver | Self Project

May 2025 – Jun 2025

- Modeled a virtual **Rubik's Cube (3x3)** in three distinct classes using standard data structures like **unordered map** available in C++
- Applied **OOP** principles such as **virtual functions**, **inheritance**, and **operator overloading** to enhance code modularity and reusability
- Achieved a **sub-3-second** solving time for a Rubik's Cube **jumbled 8 times**, applying **BFS**, **DFS**, and **Iterative Deepening DFS** algorithms
- Implemented **Korf's Iterative Deepening A* Algorithm**, achieving a **sub-10-second** solving time for a Rubik's Cube scrambled **13 times**

Vision-Guided Behavioral Monitoring System for Exam Surveillance | Self Project

May 2024

- Engineered a real-time cheat detection pipeline using **YOLOv8** and **dlib's 68-point model** to recognize mobile usage and head orientation
- Tracked pupil and facial orientations utilizing **68 facial landmarks** to identify prolonged eye diversions and unnatural visual behavior
- Achieved **86.9% mAP@50** by fine-tuning the YOLOv8 model on the **Roboflow dataset**, integrated alert generation with frame capture
- Processed continuous IP camera video stream using **OpenCV** and **multithreaded processing** via Python for seamless visual monitoring

Depth Estimation with Parallel Stereo Vision System | Self Project

May 2024

- Estimated depth in real-world conditions by replacing **LiDAR** with two calibrated, parallel monocular cameras for stereo vision analysis
- Designed a custom **depth estimation** pipeline from scratch without relying on the standard **image processing** libraries and frameworks
- Implemented a **template-matching algorithm** to calculate stereo **disparity** by identifying corresponding features between image frames
- Calculated accurate depth values using the **disparity map** and known camera intrinsics, generating a detailed **depth map** representation

COMPETITIONS

NK Securities: Research Hackathon 2025

Jun 2025

- Secured **Global Rank: 197/6000+** by modeling per-second implied volatility for NIFTY50 options, achieving **public MSE of 0.0000198**
- Engineered a high-frequency volatility forecasting pipeline using more than **178,000 actual market records** and **90+ crafted features**

Convolve 3.0: Pan-IIT AI/ML Hackathon | IIT Guwahati

Jan 2025

- Secured **Top 50 rank among 4,000+ teams** in a national-level Pan-IIT hackathon by implementing advanced machine learning algorithms
- Engineered a loan default prediction model leveraging **XGBoost** through **hyperparameter optimization** and **class imbalance techniques**

SCHOLASTIC ACHIEVEMENTS

- Ranked in the **top 1%** in the **Joint Entrance Examination (JEE) Main**, securing a rank out of over 1.2M candidates 2023
- Ranked in the **top 5%** in the **Joint Entrance Examination (JEE) Advanced**, securing a rank among 200K candidates 2023
- Earned a merit-based branch change by ranking in the **top 10%** out of over **1,800** first-year undergraduate students 2024

SKILLS AND EXPERTISE

Programming Languages: C++, C, Python, SQL **Frameworks/Packages:** NumPy, TensorFlow, OpenCV, ROS, OpenAI Gym

Developer Tools: Git, GitHub, Visual Studio Code, Jupyter Notebooks, Google Colaboratory, Anaconda, Gazebo, SolidWorks

COURSEWORK INFORMATION

Mathematics and Computer Science: Programming and Data Structures (Theory and Laboratory), Linear Algebra, Advanced Calculus, Transform Calculus, Numerical and Complex Analysis, Computer-Aided Process Engineering, Computer Graphics

MOOCs: AlgoZenith Premium, Supervised Machine Learning, Neural Networks and Deep Learning, Improving Deep Neural Networks, Structuring Machine Learning Projects, Convolutional Neural Networks, Object-Oriented Programming